PBL Phase 2 Report

End to End Pre Loan Default Assessment System

Project Id: AIML-2

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Index

S.no.	Title	Page
1.	Problem Description 1.1. Problem Description 1.2 Flow Diagram	3-4
2.	Modules for project implementation 2.1. Modules Used 2.2. Platform Used	5-7
3.	Machine Learning Models used	8
4.	Evaluation metrics used	9
5.	Results and Visualizations	10-11
6.	Conclusion and Future Scope	12
7.	References	12

1. Problem Description

The financial sector faces significant risks from 'credit risk,' where borrowers fail to repay loans, leading to substantial economic losses. This project aims to develop a predictive model that identifies potential 'loan defaulters' based on historical customer banking data. By leveraging classification algorithms, banks can automate the risk assessment process, ensuring more objective and data-driven lending decisions. In real life, this minimizes non-performing assets (NPAs) and helps maintain the stability of the banking ecosystem.

1.1. Dataset description

The dataset contains historical banking records including Age, Income, Loan Amount, Credit Score, Months Employed, and Loan Term. It includes categorical data like Education, Marital Status, and Loan Purpose.

LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	Education	EmploymentType	MaritalStatus	HasMortgage	HasDependents	LoanPurpose	HasCoSigner	Default
I38PQUQS96	56.0	85994.0	50587.0	520.0	80.0	4.0	15.23	36.0	0.44	Bachelor's	Full-time	Divorced	Yes	Yes	Other	Yes	0
HPSK72WA7R	69.0	50432.0	124440.0	458.0	15.0	1.0	4.81	60.0	0.68	Master's	Full-time	Married	No	No	Other		0
C1OZ6DPJ8Y	46.0	84208.0	129188.0	451.0	26.0	3.0	21.17	24.0	0.31	Master's	Unemployed	Divorced	Yes	Yes	Auto	No	1
V2KKSFM3UN	32.0	31713.0	44799.0	743.0	0.0	3.0	7.07	24.0	0.23	High School	Full-time	Married	No	No	Business	No	0
EY08JDHTZP	60.0	20437.0	9139.0	633.0	8.0	4.0	6.51	48.0	0.73	Bachelor's	Unemployed	Divorced	No	Yes	Auto	No	0
A9S62RQ7US	25.0	90298.0	90448.0	720.0	18.0	2.0	22.72	24.0	0.1	High School	Unemployed	Single	Yes	No	Business	Yes	1
H8GXPAOS71	38.0	111188.0	177025.0	429.0	80.0	1.0	19.11	12.0	0.16	Bachelor's	Unemployed	Single	Yes	No	Home	Yes	0
0HGZQKJ36W	56.0	126802.0	155511.0	531.0	67.0	4.0	8.15	60.0	0.43	PhD	Full-time	Married	No	No	Home	Yes	0
1R0N3LGNRJ	36.0	42053.0	92357.0	827.0	83.0		23.94	48.0	0.2	Bachelor's	Self-employed	Divorced	Yes	No	Education	No	1
CM9L1GTT2P	40.0	132784.0	228510.0	480.0	114.0	4.0	9.09	48.0	0.33	High School	Self-employed	Married	Yes	No	Other	Yes	0
1A3XVH6ZO	28.0	140466.0	163781.0	652.0	94.0	2.0	9.08	48.0	0.23	High School	Unemployed	Married	No	No	Education	No	0
Y8UETC3LSG	28.0	149227.0	139759.0	375.0	56.0	3.0	5.84	36.0	0.8	PhD	Full-time	Divorced		No	Education	Yes	1
RM6QSRHIYP	41.0	23265.0	63527.0	829.0	87.0	4.0	9.73	60.0	0.45	Master's	Full-time	Divorced	Yes	No	Auto	Yes	0
GX5YQOGROM	53.0	117550.0	95744.0	395.0	112.0	4.0	3.58	24.0	0.73	High School	Unemployed	Single	No	No	Auto	Yes	0
X0BVPZLDC0	57.0	139699.0	88143.0	635.0	112.0	4.0	5.63	48.0	0.2	Master's		Divorced	No	No	Home	No	0
O5DM5MPPNA	41.0	74064.0	230883.0	432.0	31.0	2.0	5.0	60.0	0.89	Master's	Unemployed	Married	Yes	No	Auto	No	0
ZDDRGVTEXS	20.0	119704.0	25697.0	313.0	49.0	1.0	9.63	24.0	0.28	PhD	Unemployed	Single	Yes	No	Home	No	0
9V0FJW7QPB	39.0	33015.0	10889.0	811.0	106.0	2.0	13.56	60.0	0.66	Master's	Self-employed	Single	Yes	No	Other	No	0
O1IKKLC69B	19.0	40718.0	78515.0	319.0	119.0	2.0	14.0	24.0	0.17	Bachelor's	Self-employed	Divorced	Yes	No	Education	No	1
F7487UU2BF	41.0	123419.0	161146.0	376.0		4.0	16.96	60.0	0.39	High School	Self-employed	Single	Yes	No	Other	Yes	0
7ASF0IHRIT	61.0	30142.0	133714.0	429.0	96.0	1.0	15.58		0.65	PhD	Part-time	Divorced	No	Yes	Business	No	0
A22KI186SE	47.0	146113.0	100621.0	419.0	55.0	1.0	9.32	12.0	0.38	Bachelor's	Unemployed	Married	Yes	Yes	Business	No	0
1HUSHWD9TW	55.0	132058.0	130912.0	583.0	48.0	4.0	5.82	60.0	0.47	High School	Unemployed	Married	No	Yes	Business	Yes	0
LXK7UEMLX0	19.0	118989.0	123300.0	528.0	73.0	3.0	15.29	36.0	0.22	PhD	Part-time	Single	Yes	No	Business	Yes	1
995RE1TIB4	38.0	56848.0	168918.0	468.0	73.0	1.0	19.1	24.0	0.22	Bachelor's	Unemployed	Single	No	No		No	0
D17PDP8LBL	50.0	81649.0	78193.0	839.0	110.0	1.0	21.41	48.0	0.5	Master's	Part-time	Married	Yes	No	Business	Yes	0
C35RYEXWJ0	29.0	114651.0	197648.0	343.0	58.0	3.0	21.07	24.0	0.19	Bachelor's	Part-time		Yes	No	Home	Yes	0
G8AIMXSES2	39.0	17633.0	167105.0	514.0	62.0	3.0	7.86	36.0	0.66	High School	Full-time	Single	Yes	Yes	Auto	Yes	1
BJNLQ0H95H	61.0	62519.0	29676.0	462.0	16.0	1.0	23.91	48.0	0.12	Bachelors	Unemployed	Divorced	Yes	No	Home	Yes	0
YIGLPWKNH5	42.0	141412.0	197764.0	580.0	57.0	2.0	10.18	12.0		Bachelor's	Full-time	Married	No	No	Education		0
GAA8OQN796	66.0	39568.0	58945.0	604.0	37.0	4.0	6.67	12.0	0.1	High School	Unemployed	Divorced	Yes	Yes	Auto	Yes	0
P3EX8G0AYT	44.0	100284.0	225403.0	551.0	31.0	1.0	18.77	36.0	0.17	Master's	Unemployed	Divorced	No		Business	Yes	1
KD97QJJFD8	59.0	102292.0	55337.0	840.0	6.0	1.0		60.0	0.44	Master's	Unemployed	Married	Yes	No	Auto	No	0
O8G74YT5W3	45.0	85673.0		787.0	103.0	4.0	22.42	24.0	0.82	PhD	Full-time	Divorced	No	No		Yes	0
1CW00BWTLF	33.0	92448.0	66282.0	607.0	39.0	1.0	11.31	12.0	0.43	Bachelor's	Part-time	Divorced	Yes	No	Home	Yes	0
PBQ09B6L9D	32.0	102178.0	179279.0	669.0	51.0	2.0	14.98	12.0	0.78	Master's	Self-employed	Single	Yes	No	Home	Yes	0
BNTWNUHHTY	64.0	102463.0	218433.0	506.0	24.0	2.0	9.23	60.0	0.86	Master's	Unemployed	Married		No	Auto	Yes	0
ASYFXCP452	68.0	85409.0	44772.0	540.0	105.0	1.0	2.96	36.0	0.17	Master's	Unemployed	Divorced	No	No	Auto	No	0
4857M8R0Y1	61.0	26470.0	19818.0	695.0	47.0	2.0	20.0	36.0	0.69	PhD	Unemployed	Divorced	Yes	Yes	Auto	Yes	0
5FENBP2UV8	69.0	87295.0	16281.0	707.0	94.0	1.0	13.82	60.0	0.75	Bachelor's	Part-time	Single	No		Other	No	0
EGBQ6R80VB	20.0	139321.0	43049.0	458.0	117.0	2.0	7.16	36.0	0.53	Master's	Self-employed	Single	Yes	No	Home		1
RT511ZZNF2	54.0		84115.0	386.0	32.0	2.0	20.89	60.0	0.78	PhD	Unemployed	Married	Yes	Yes	Business	Yes	0
KAC7P2RE1X	68.0	111716.0	215851.0	747.0	99.0	3.0	18.84	60.0	0.4	High School		Divorced	No	Yes	Home	No	0
XGSWPXX0TY	24.0	105732.0	33013.0	400.0	58.0	3.0	15.64	36.0	0.48	High School		Married	Yes	Yes	Auto	Yes	0
2XUD7N4OJ1	38.0	73117.0	221628.0	639.0	84.0	4.0	13.05	36.0	0.56	Bachelor's	Part-time	Married	No	Yes	Home	Yes	0
2O7VM6EN0D	26.0	125268.0	67671.0	795.0	72.0	3.0	11.42	12.0	0.3	High School	Self-employed	Single	No	No	Business	No	0
Z7UFZIW3MK	56.0	24796.0	37657.0	498.0	88.0	1.0	8.2		0.73	High School	Part-time	Married	Yes	Yes	Auto	No	0
RSP1YD80Z7	35.0	95963.0	77552.0	560.0	8.0	2.0	6.63	24.0	0.86	Master's	Self-employed	Divorced		Yes	Home	No	1
AFJTOPMFJV	21.0	16170.0	66883.0	505.0	27.0	1.0	4.41	48.0	0.42	PhD	Self-employed	Single	No	Yes	Education	No	0
LHQOFSPTTT	42.0	143244.0	240591.0	393.0	96.0	3.0	23.15	36.0	0.16	High School	Self-employed	Married	No	Yes	Business	No	0
06Q27SJKQB	31.0	103515.0	220032.0	369.0	57.0	2.0	22.36	24.0	0.81	phd	Unemployed	Divorced	No	No	Education	Yes	1
N3KDA4UM9K	67.0	100949.0	145906.0	439.0	36.0	2.0	3.74	48.0	0.58	Bachelor's	Unemployed	Married	No	No	Auto	Yes	0
130U8TZ0HW	26.0	92153.0	214064.0	634.0	33.0	3.0	2.34	24.0	0.32	PhD	Full-time	Married	No	Yes	Education	No	0
E9NX4IRVSU	43.0	70113.0	215064.0	657.0	0.0	2.0	6.0	24.0	0.76	Bachelor's		Single	No	Yes	Home	Yes	0
963HFKM3CT	19.0	24611.0	41605.0	417.0	59.0	3.0	15.11	48.0	0.59	Bachelor's	Unemployed	Single	No	Yes	Business	No	0
QHPWTGMTU8	37.0	37173.0	234059.0	455.0	20.0	3.0	15.12	48.0	0.74	PhD	Full-time	Married	No	Yes	Other	No	0
0TS3GR64AD	45.0	31137.0	158128.0	351.0	54.0	3.0	5.4	24.0	0.25	Bachelor's	Unemployed	Single	No	No	Other	Yes	0
DSOB0M5AQ4	64.0	49565.0	245711.0	655.0	76.0	4.0	23.41	24.0	0.37	High School	Unemployed	Single	No	Yes	Auto	Yes	0
PWVQRNYJGA	24.0	31644.0	75317.0	737.0	66.0	4.0	5.65	36.0	0.51	High School	Full-time	Divorced	No	No	Business	Yes	0
0VEK3DDM69	61.0	49113.0	222046.0	771.0	47.0	3.0	22.8	12.0	0.24	High School	Unemployed	Divorced	No	Yes	Other	No	0
LCBMJZGE8D	25.0	85814.0	227368.0	669.0	58.0	1.0	19.11	36.0	0.75	High School	Unemployed	Divorced	Yes	No	Education	Yes	1
Z3WJUM1DZ	64.0	110606.0	137139.0	669.0	11.0	3.0	18.01	60.0	0.66	Master's	Part-time	Married	No	Yes	Home		0
YQKG397DX3	52.0	76136.0	145318.0	775.0	23.0	3.0	12.44	24.0	0.15	PhD	Unemployed	Single	No	Yes	Home	No	0
HRVBDX8JP7	31.0	82461.0	222343.0	680.0	22.0	1.0		24.0	0.36	Master's	Unemployed	Divorced	No	No	Home	No	0
XPEW5QYJ3S	34.0	37082.0	175274.0	694.0	67.0	3.0	15.59	36.0	0.38	Bachelor's	Unemployed	Single	Yes	No	Business	No	0
092M7ZSZ6M	53.0	149401.0	245626.0	534.0	18.0	1.0	15.82	60.0	0.25	Bachelor's	Unemployed	Divorced	No	Yes	Auto	No	0
ZOPZLF57NR		77135.0	159839.0	529.0	59.0	4.0	4.14	36.0	0.41	High School	Full-time	Married	Yes	Yes	Other	No	0
648L7QUFVO	57.0	34830.0	63072.0	416.0	93.0	3.0	20.34	60.0	0.45	Bachelor's	Unemployed	Married	No	No	Home	No	0

Fig 1: Sample dataset

1.2. Flow diagram

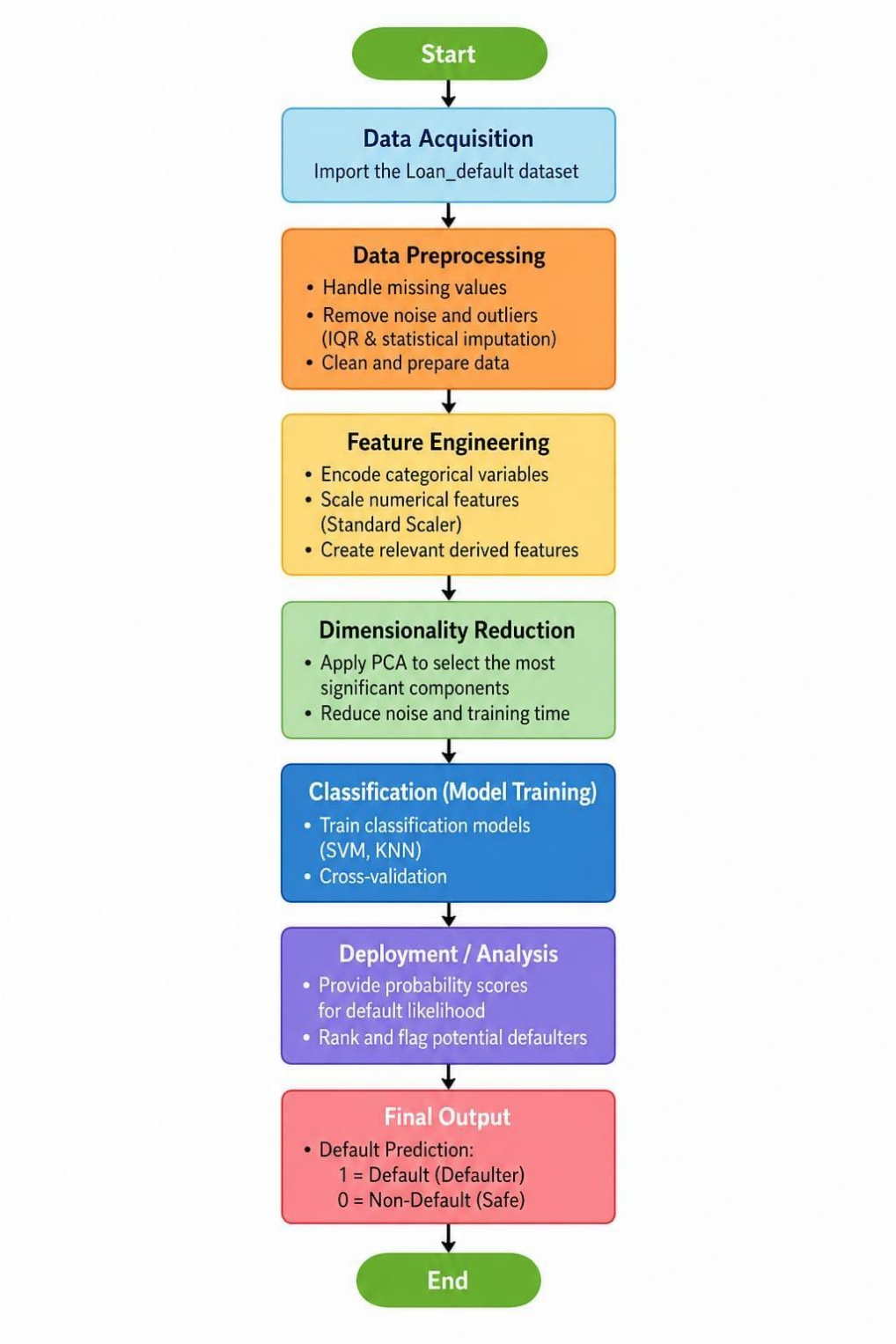

Fig 2: Flow diagram of the proposed system.

2. Modules for Project Implementation

2.1 Modules used

Pandas

Pandas acts as the backbone of the project's data management. In this system, it is used to load the Loan_default.csv file into a DataFrame, allowing for easy manipulation of rows and columns. Beyond just reading data, it is used to perform feature selection and data cleaning, such as identifying null values and calculating statistical summaries like the mean or median of borrower incomes. It essentially transforms raw, messy banking data into a structured format ready for analysis.

Numpy

NumPy is used for the underlying mathematical computations that drive the machine learning models. While Pandas handles the tables, NumPy handles the multi-dimensional arrays and complex math, such as calculating distance metrics for the KNN algorithm. It is essential for performing element-wise operations on large datasets, ensuring that heavy calculations are performed with high speed and efficiency, which is critical when dealing with thousands of customer records.

Scikit-Learn (Preprocessing)

This module is responsible for "normalizing" the data so that the models can interpret it fairly. The StandardScaler is used to ensure that features with large ranges (like Income) don't overrule smaller numerical features during model training. It also includes LabelEncoder, which converts categorical text—such as "Full-time" or "Unemployed"—into numbers that the SVM and Random Forest algorithms can process. Without this, the models would be unable to mathematically compare different types of borrower information.

Imbalanced-Learn (SMOTE)

In credit risk datasets, "Defaulters" are typically a small percentage of the total data. If trained on this unbalanced data, a model will likely just learn to predict "Non-Default" for everyone. SMOTE (Synthetic Minority Over-sampling Technique) solves this by creating "synthetic" examples of defaulters. By balancing the dataset, the models learn the specific characteristics of high-risk borrowers more effectively, significantly improving the Recall metric of the system.

Matplotlib & Seaborn

These are the primary tools for Exploratory Data Analysis (EDA). Seaborn Heatmaps are used to visualize the correlation matrix, allowing the bank to see how strongly a "Credit Score" relates to a "Default." Histograms display the distribution of loan amounts, while Boxplots help identify "Outliers"—such as borrowers with impossibly high incomes—which might indicate data errors. These visualizations are crucial for communicating findings to stakeholders.

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2.2. Platform used

Jupyter Notebook

• **Interactive Development:** Jupyter Notebook provided an interactive environment that allowed for executing credit risk analysis code in "cells," making it easy to test individual preprocessing steps like SMOTE or scaling without rerunning the entire script.

• **Integrated Data Visualization:** It enabled the direct display of Seaborn heatmaps and Matplotlib histograms within the workspace, which was essential for Exploratory Data Analysis (EDA) to visualize correlations between credit scores and defaults.

• **Literate Programming:** The platform allowed for the combination of executable Python code with Markdown documentation, enabling the team to explain the logic behind model choices (like SVM vs. KNN) alongside the actual implementation.

• **Sequential Data Workflow:** Its linear structure perfectly matches the machine learning pipeline of this project—moving systematically from data loading and cleaning to dimensionality reduction (PCA) and final model evaluation.

• **Debugging and Optimization:** Jupyter's ability to maintain the state of variables in memory allowed for fine-tuning Random Forest hyperparameters and re-evaluating metrics like "Recall" instantly, ensuring the most accurate risk assessment results.

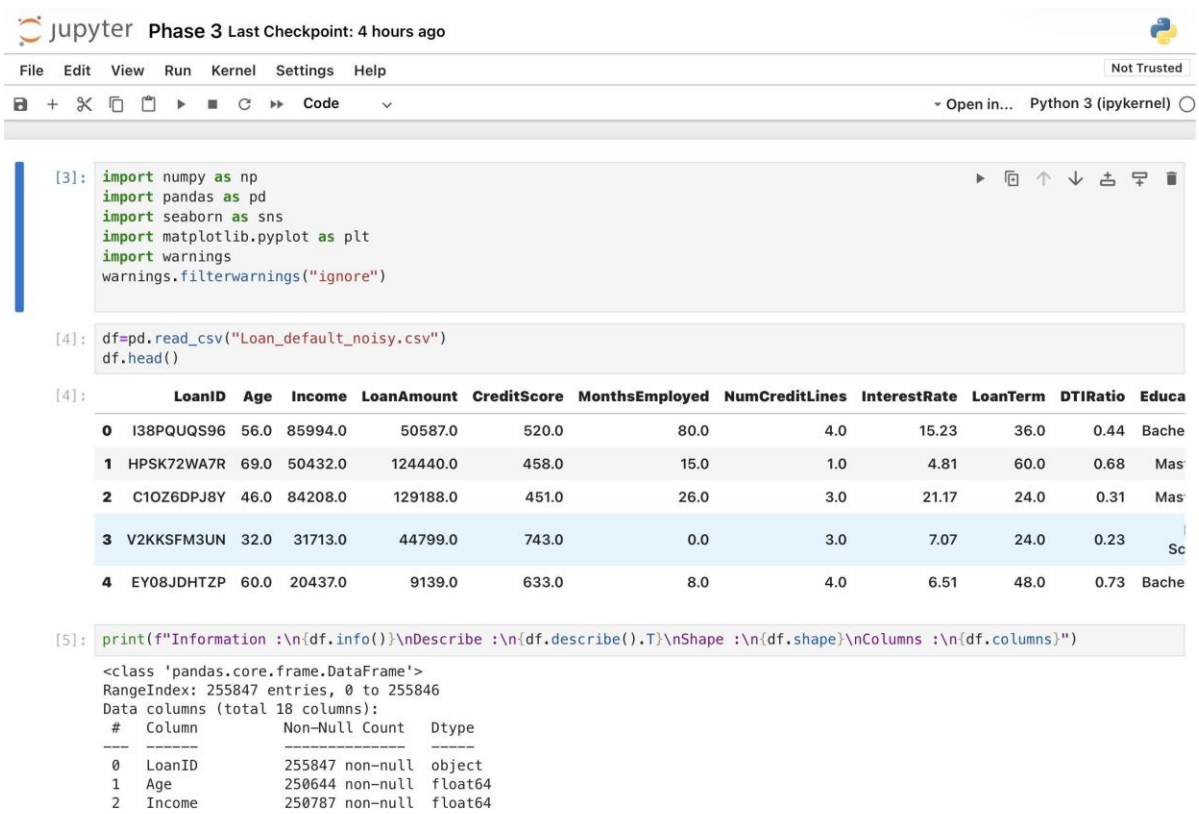

Fig 3: Jupyter Notebook

3. Machine learning Models Used

Support Vector Machines (SVM)

SVM is used to find the optimal hyperplane that maximizes the margin between classes (Default vs. Non-Default). We utilize various kernels such as Linear, Poly, and RBF to handle non-linear relationships in the banking data.

K-Nearest Neighbors (KNN)

KNN classifies a borrower based on the characteristics of the 'K' most similar historical customers. It is effective for identifying patterns in borrower behavior without assuming a specific data distribution.

Principal Component Analysis (PCA)

An unsupervised model used to compress the feature space. It identifies the principal components that explain the maximum variance in the banking dataset, helping the classifiers run faster and potentially avoid overfitting.

Logistic Regression (Planned/Baseline)

While the core analysis focuses on SVM and KNN, the reports and synopsis mention Logistic Regression as a baseline model. This is a statistical model used for binary classification. It estimates the probability of an event (default) occurring based on input features using a logistic function.

Random Forest Classifier

Random Forest is an "Ensemble Learning" method that operates by constructing a multitude of Decision Trees during training. It makes a final prediction by taking the majority vote (for classification) from all the individual trees.

4. Evaluation Metrics Used

- **Accuracy**: Percentage of total correct predictions.

- **Recall:** Critical in banking; it measures the ability to identify all actual defaulters to minimize losses.

- **F1-Score:** Provides a balance between Precision and Recall, ensuring the model isn't biased.

- **Confusion Matrix:** Visualizes the performance of the classification algorithm. A confusion matrix shows:
- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

Importance:

Helps analyze model performance

Identifies misclassification

- **Heatmap (Confusion Matrix Visualization)**: A heatmap visually represents the confusion matrix using colors.

Advantages:

Easy interpretation

Detects errors quickly

5. Results and Visualizations

Initial Exploratory Data Analysis (EDA) shows that interest rates and credit scores are the most significant indicators of default. Heatmaps generated during the analysis helped identify multi-collinearity between features.

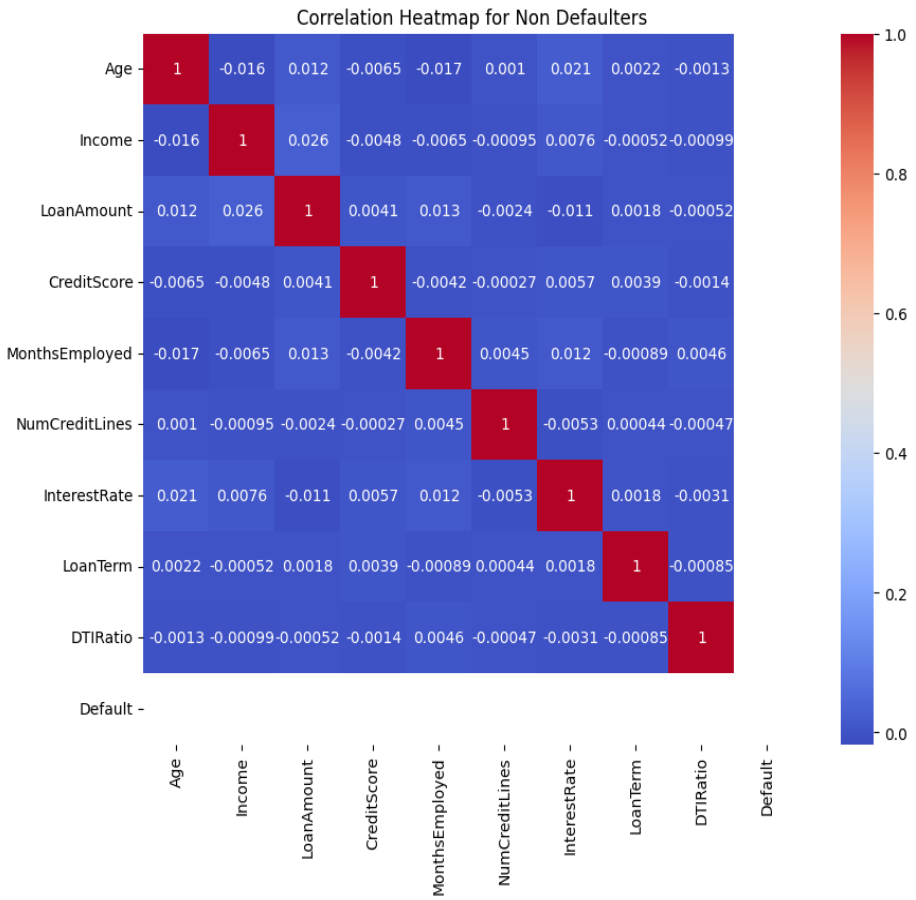

Fig 4: Figure showing Heatmap of the Feature correlation

```
f1 Score :0.7401439888730468
accracy score : 0.6854232487199041
confusion matrix :
[[11100  5099]
 [  676  1483]]
classification report :
              precision    recall  f1-score   support

         0.0       0.94      0.69      0.79     16199
         1.0       0.23      0.69      0.34      2159

    accuracy                           0.69     18358
   macro avg       0.58      0.69      0.57     18358
weighted avg       0.86      0.69      0.74     18358
```

Fig 5: Figure showing linear SVC report

```
f1 Score :0.7277073869809259
accracy score : 0.6732759559864909
confusion matrix :
[[11338  4861]
 [ 1137  1022]]
classification report :
              precision    recall  f1-score   support

         0.0       0.91      0.70      0.79     16199
         1.0       0.17      0.47      0.25      2159

    accuracy                           0.67     18358
   macro avg       0.54      0.59      0.52     18358
weighted avg       0.82      0.67      0.73     18358
```

Fig 6: Figure showing knn report

	A	B	C	D	E	F
1	Model	Accuracy	Precision (Class	Recall (Class 1)	Weighted F1-Sc	ROC-AUC
2	Voting Ensemble	0.71	0.24	0.65	0.76	0.75
3	XGBoost	0.73	0.23	0.57	0.77	0.74
4	Random Forest	0.7	0.23	0.65	0.75	0.75
5	SVM (RBF)	0.72	0.23	0.59	0.76	0.73
6	Logistic Regress	0.69	0.23	0.68	0.74	0.75
7	K-NN (Baseline)	0.67	0.17	0.47	0.73	0.57
8						

Fig 7: Figure showing information about all models used

6. Conclusion & Future Scope

• **Conclusion:** The project successfully automated credit risk assessment using ensemble learning. By balancing the data with SMOTE and using a Voting Classifier, we created a system that minimizes financial loss (NPAs) through objective, data-driven lending decisions.

• **Future Scope:** Future enhancements include deploying the model as a web application using Flask, integrating real-time transaction monitoring, and using Explainable AI to provide transparent reasons for loan rejections.

7. References

Books:

1. 1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow – Aurélien Géron
2. 2. Pattern Recognition and Machine Learning – Christopher Bishop

Online Resources:

- 1. Scikit-Learn Documentation: https://scikit-learn.org/
- 2. Pandas Official Guide: https://pandas.pydata.org/